

Sequence Diagram Guide

4/2019

- What is a Sequence Diagram?

Sequence diagrams

- Sequence diagrams are part of the UML
- used to model the interactions between the actors and the objects within a system.
- A sequence diagram shows the sequence of interactions that take place during a particular use case or use case instance.
 - The objects and actors involved are listed along the top of the diagram, with a dotted line drawn vertically from these.
 - Interactions between objects are indicated by annotated arrows.

<http://creately.com/blog/diagrams/sequence-diagram-tutorial/>

Sequence diagram for View patient information

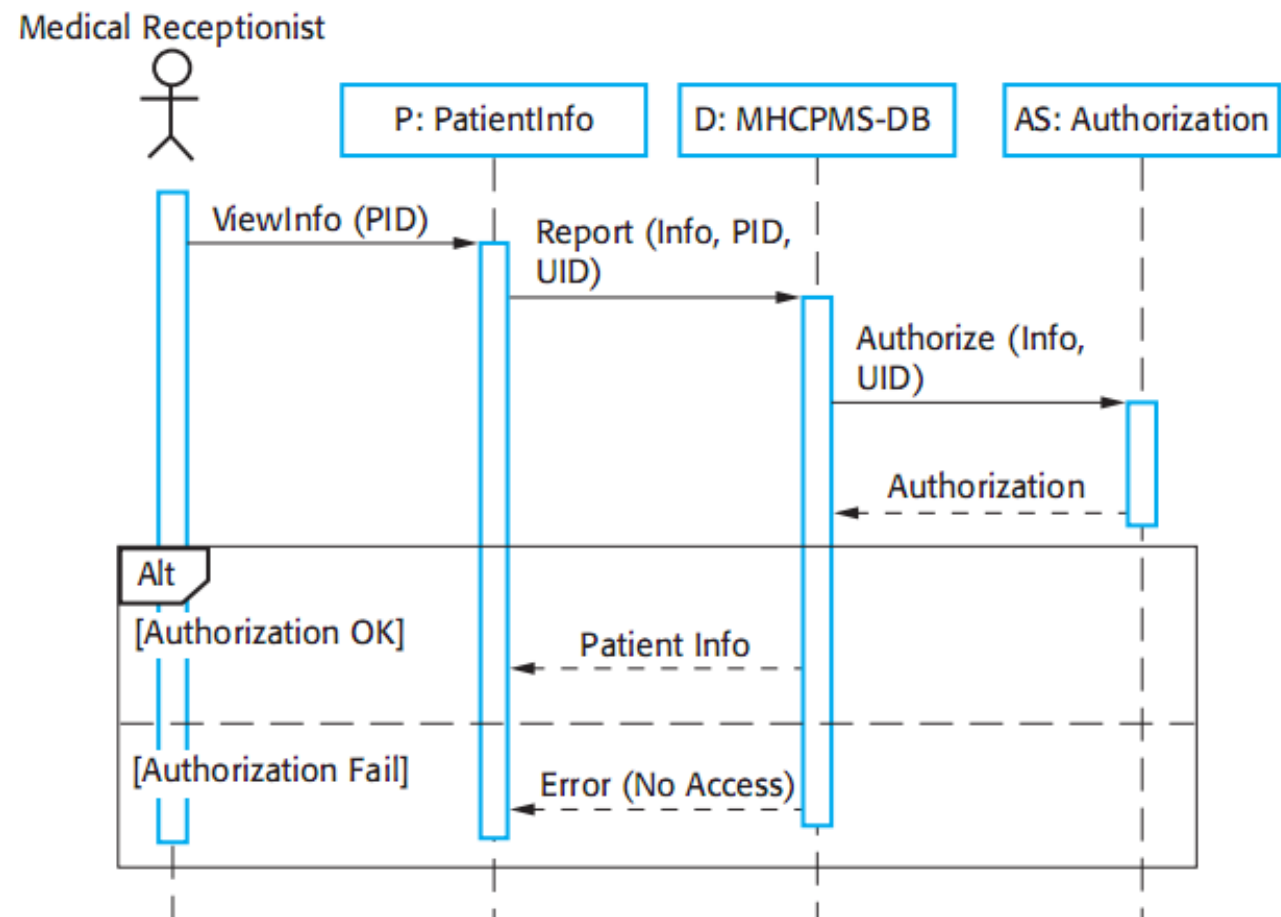
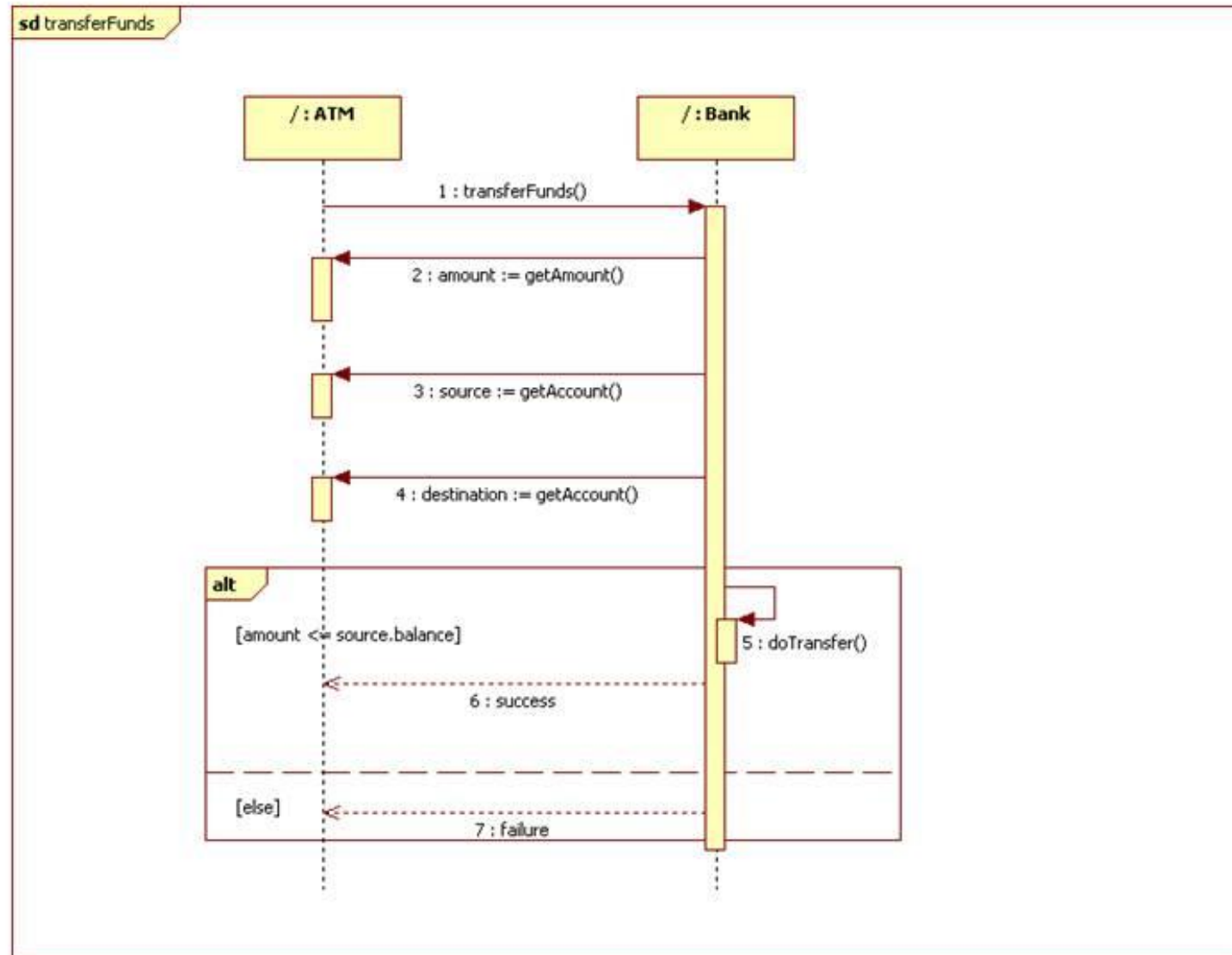


Figure 5.6 Sequence diagram for View patient information

Another sequence diagram



Sequence Diagram

- Sequence diagrams, commonly used by developers, **model the interactions between objects in a single use case?**

→ part of use case

→ no use case

→ many use case

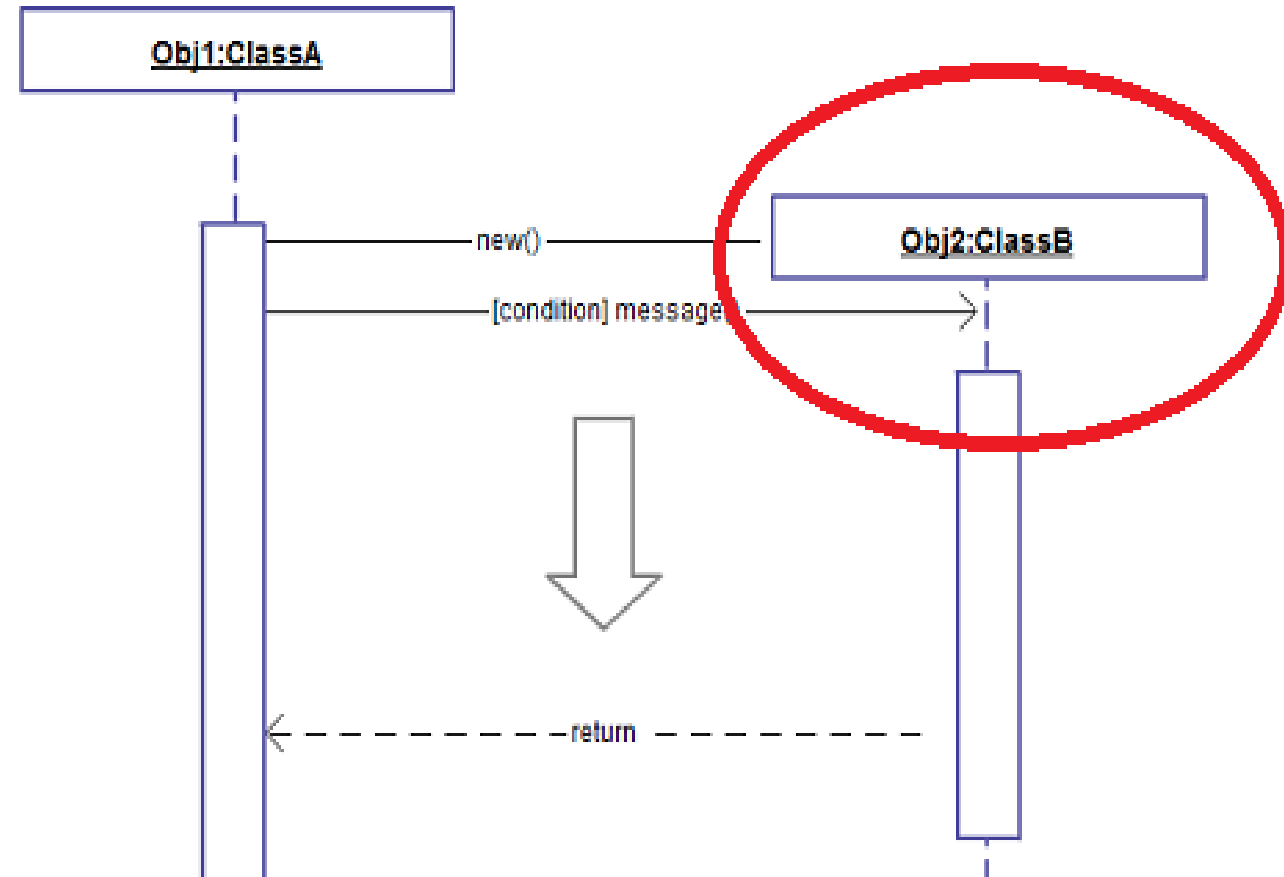
used to model the interactions between the actors and the objects **within a system.**

Getting to Know the Basics

- 1. Lifelines
- 2. Messages
- 3. Guards
- 4. Alternatives
- 5. Option
- 6. Loops
- 7. Break
- 8. Parallel
- 9. *Reference Fragment*
- ...

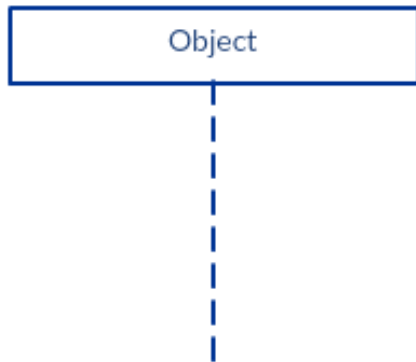
1. Lifelines

- Lifeline notation elements are placed across the top of the diagram.
- Lifelines are representative of roles or object instances that partake in the sequence being modeled.
- Lifelines are shown as a box with a dashed line descending from the center of the bottom edge.
- the lifeline represents a specific instance of a class in a sequence diagram.
- **Noun – object:Class**

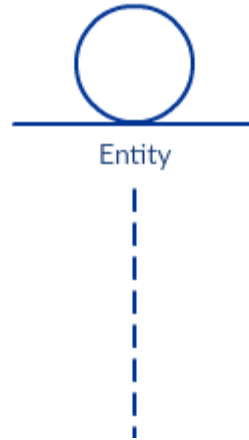


1. Lifelines

- **Lifeline Notation**



represent
system data



Ex: in a customer service application, the Customer entity would manage all data related to a customer.

indicates a system
boundary/ software
element in a system



Ex: user interface screens, database gateways or menus that users interact with, are boundaries.

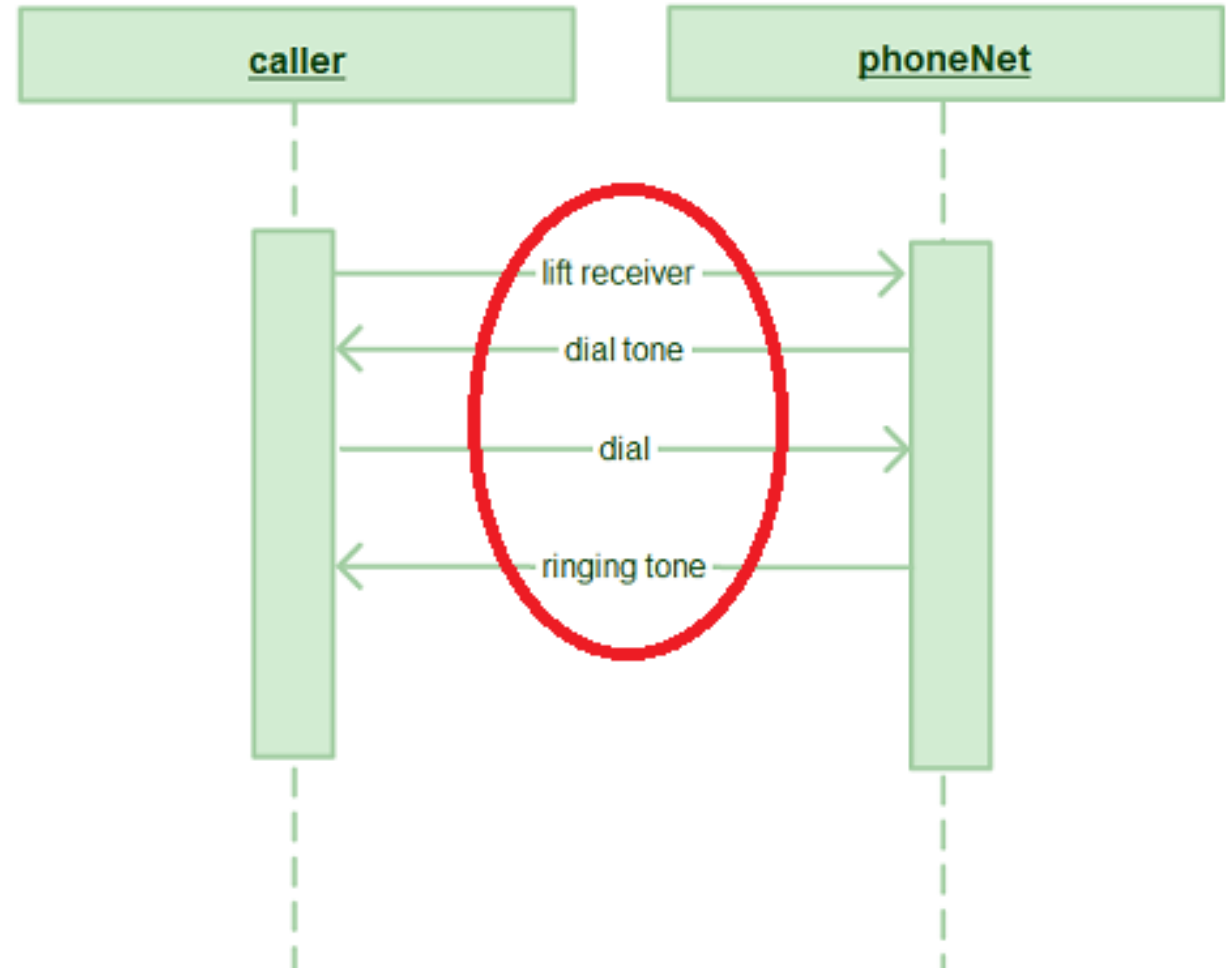
indicates a
controlling entity
or manager



organizes and schedules the interactions between the boundaries and entities and serves as the mediator between them

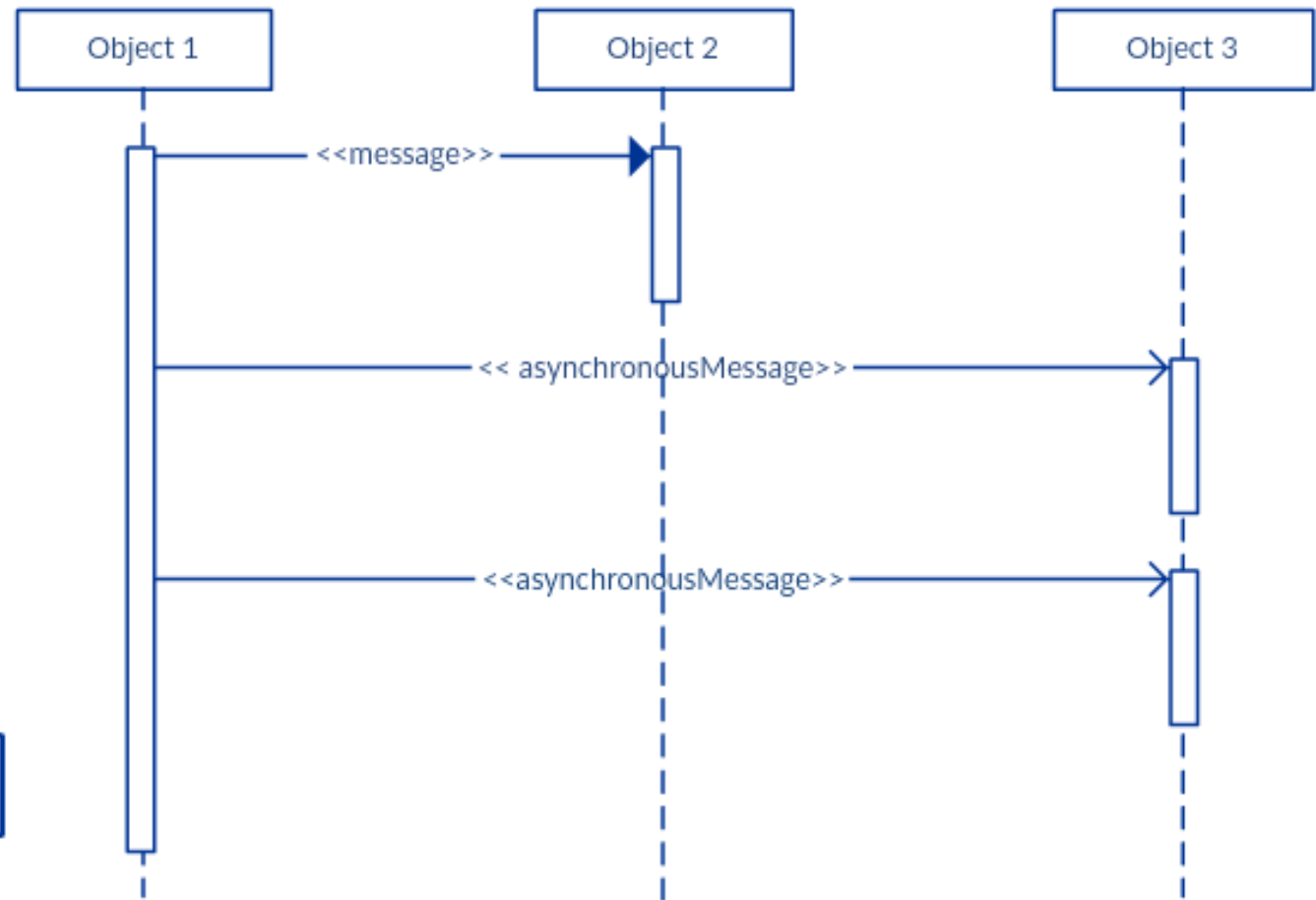
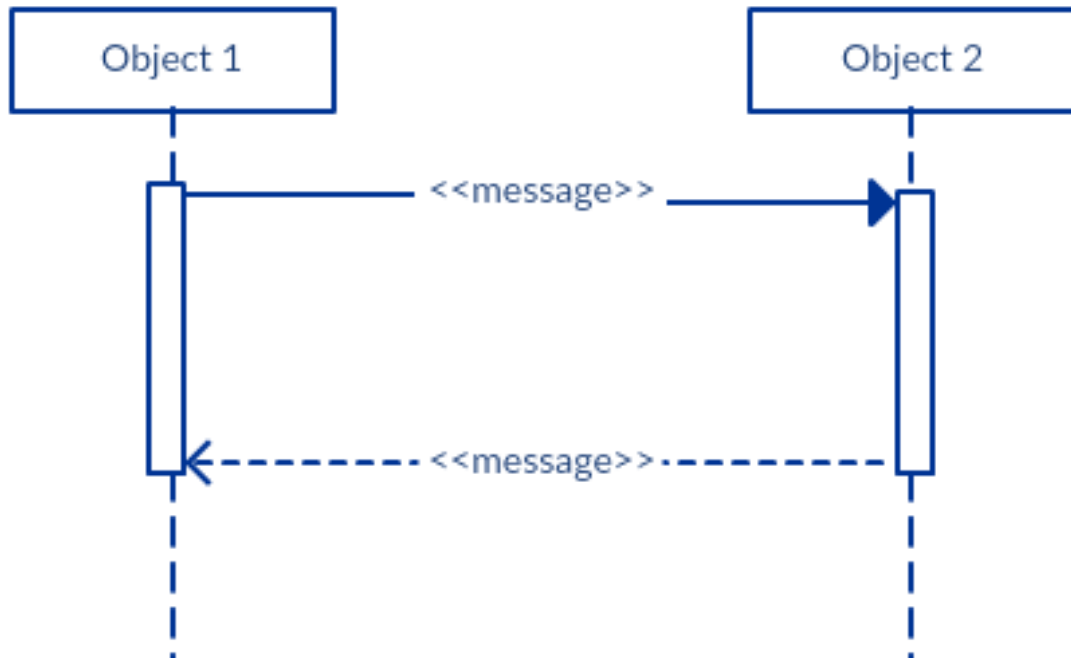
2. Messages

- To show an object or lifeline sending a message to another object, you draw a line to the receiving object
- with a solid arrowhead (if a synchronous call operation).
- with a stick arrowhead (if an asynchronous signal)
- The message/method name is placed above the arrowed line



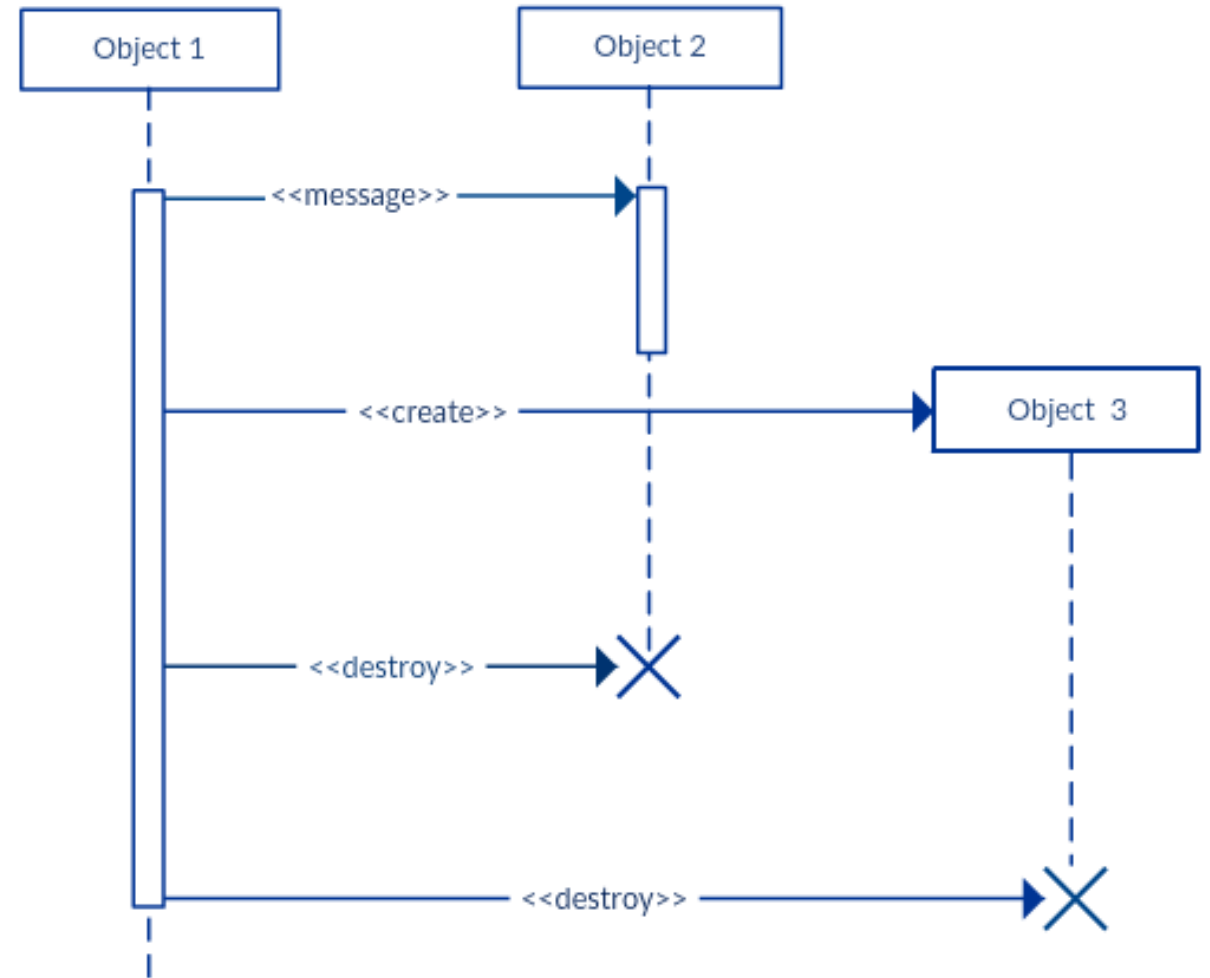
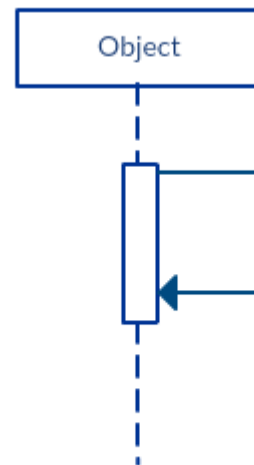
2. Messages

- *Synchronous message*
- *Asynchronous message*
- *Return message*



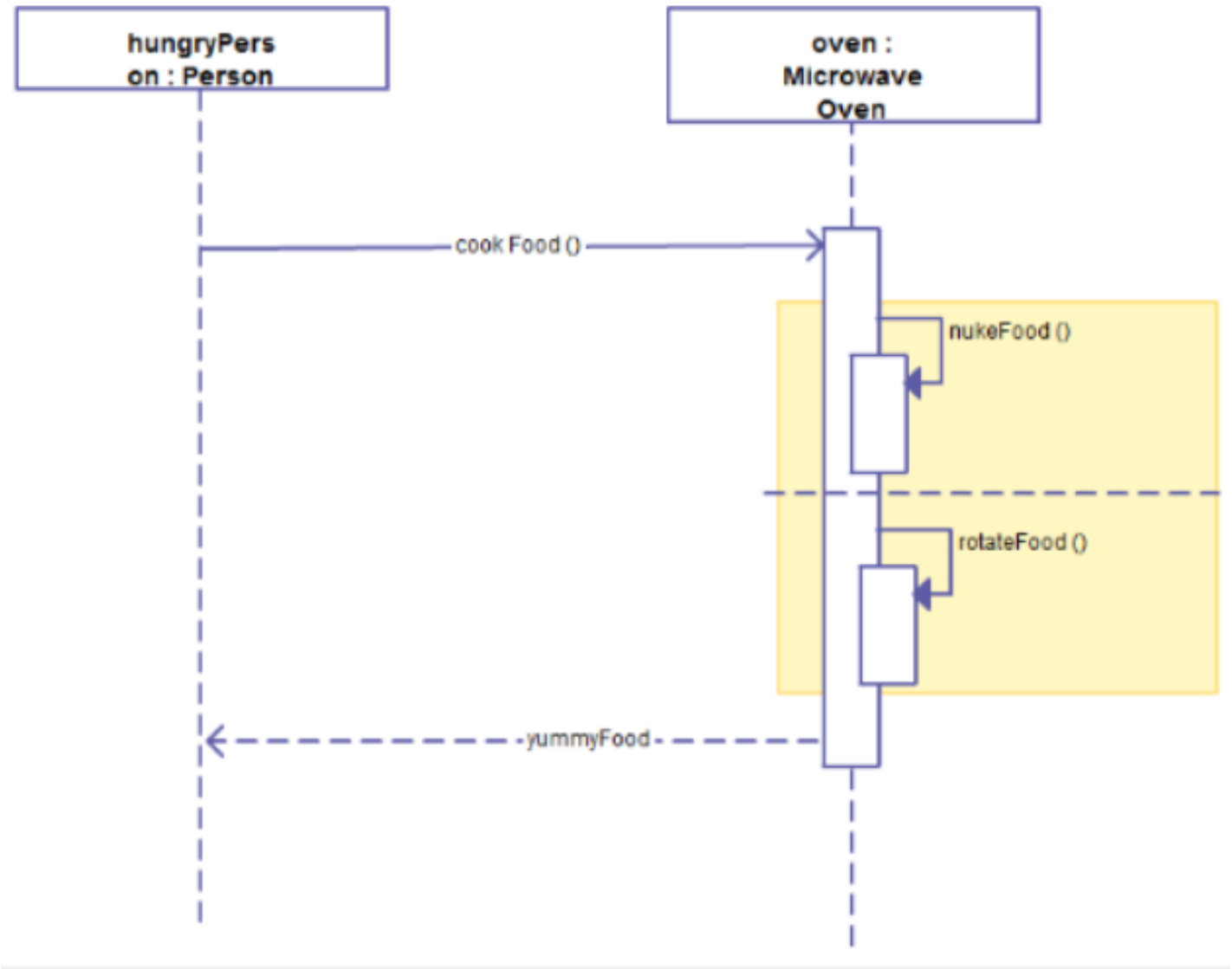
2. Messages

- *Participant creation message*
- *Participant destruction message*
- *Reflexive message*



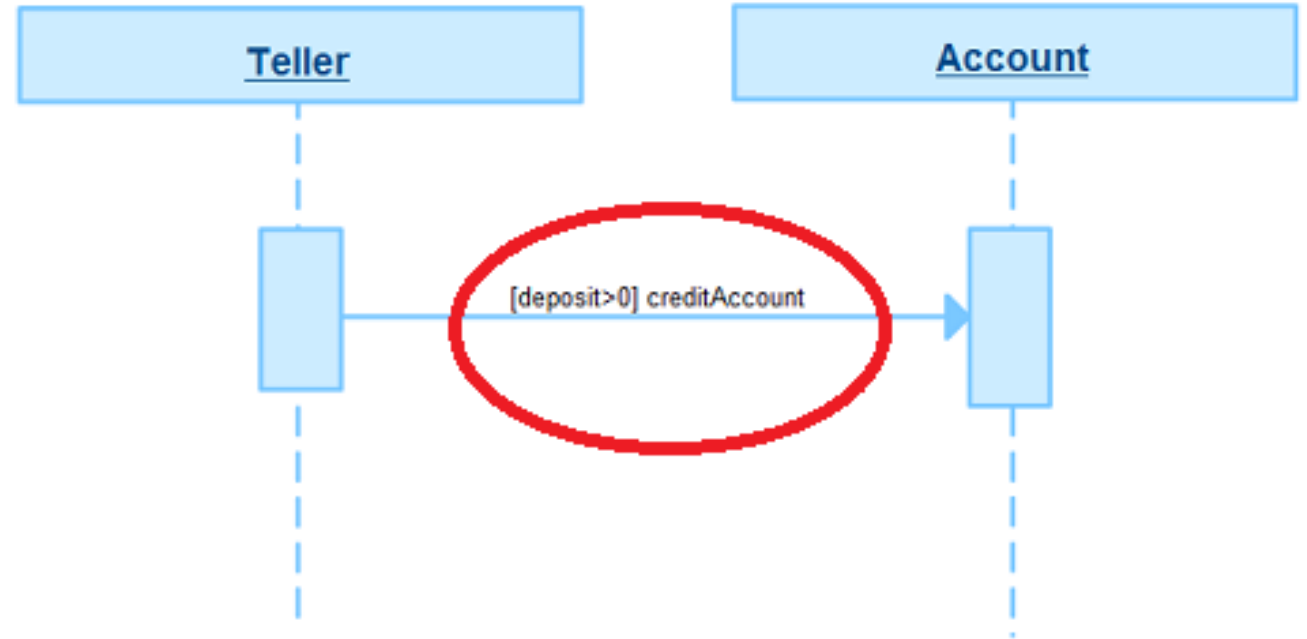
2. Messages

- *Reflexive message* example



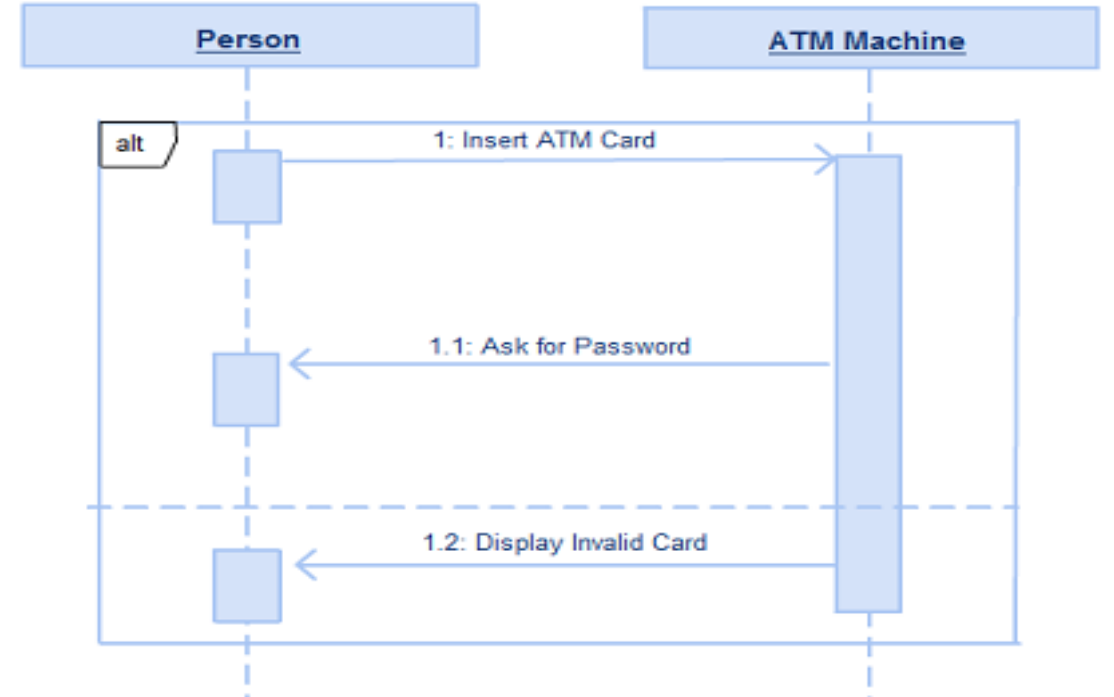
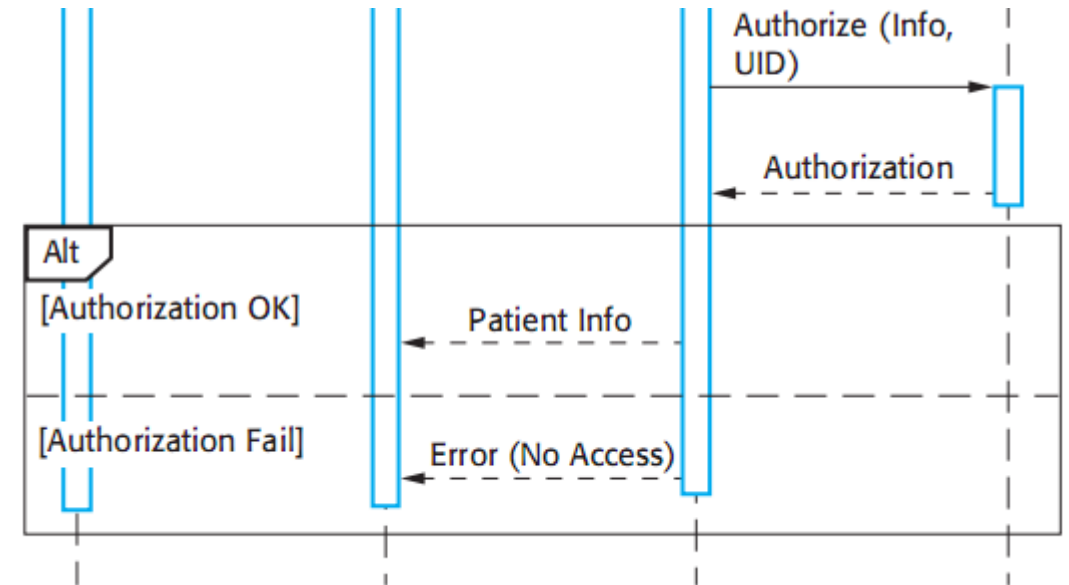
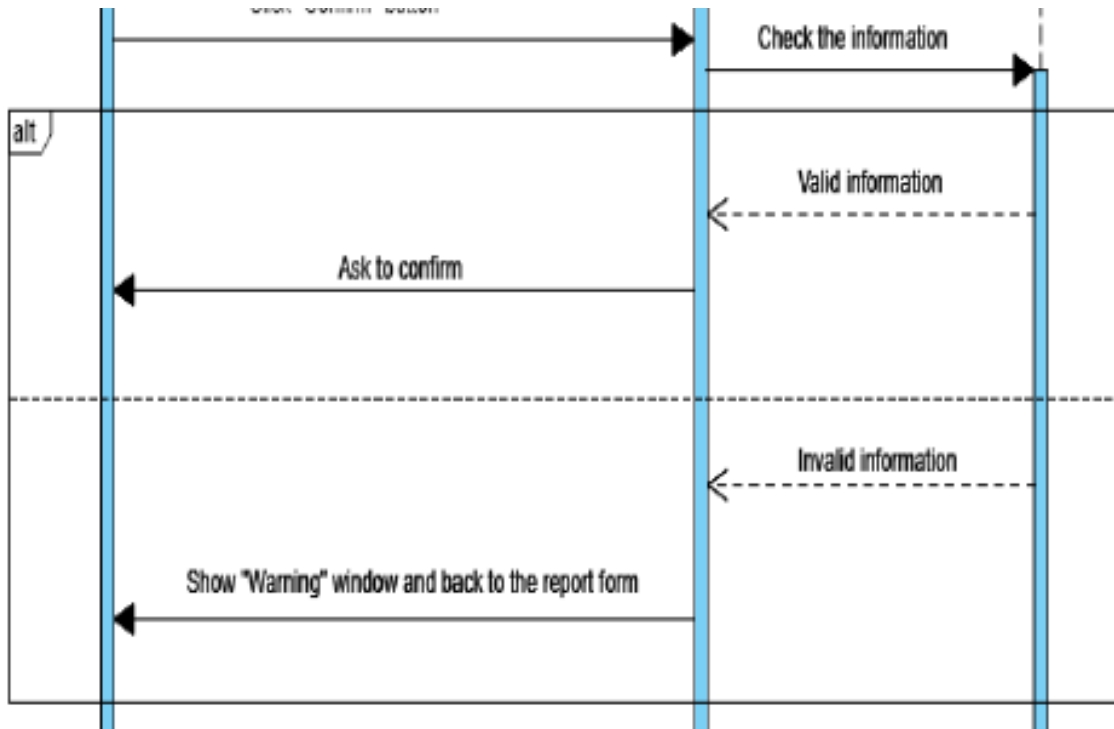
3. Guards

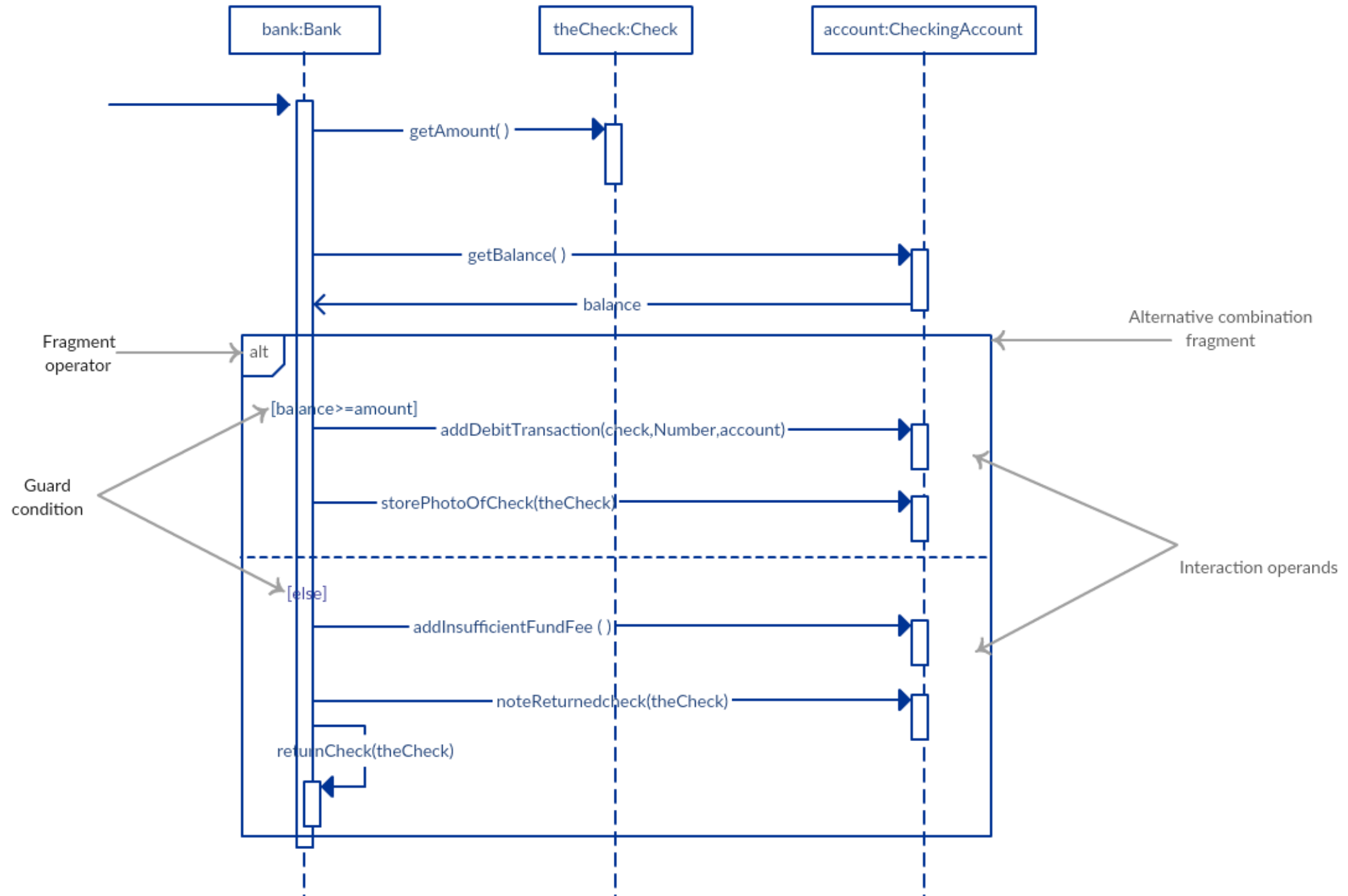
- When modeling object interactions, there will be times when a condition must be met for a message to be sent to an object.
- Guards are conditions that need to be used throughout UML diagrams to control flow
- a guard could only be assigned to a single message



4. Alternatives

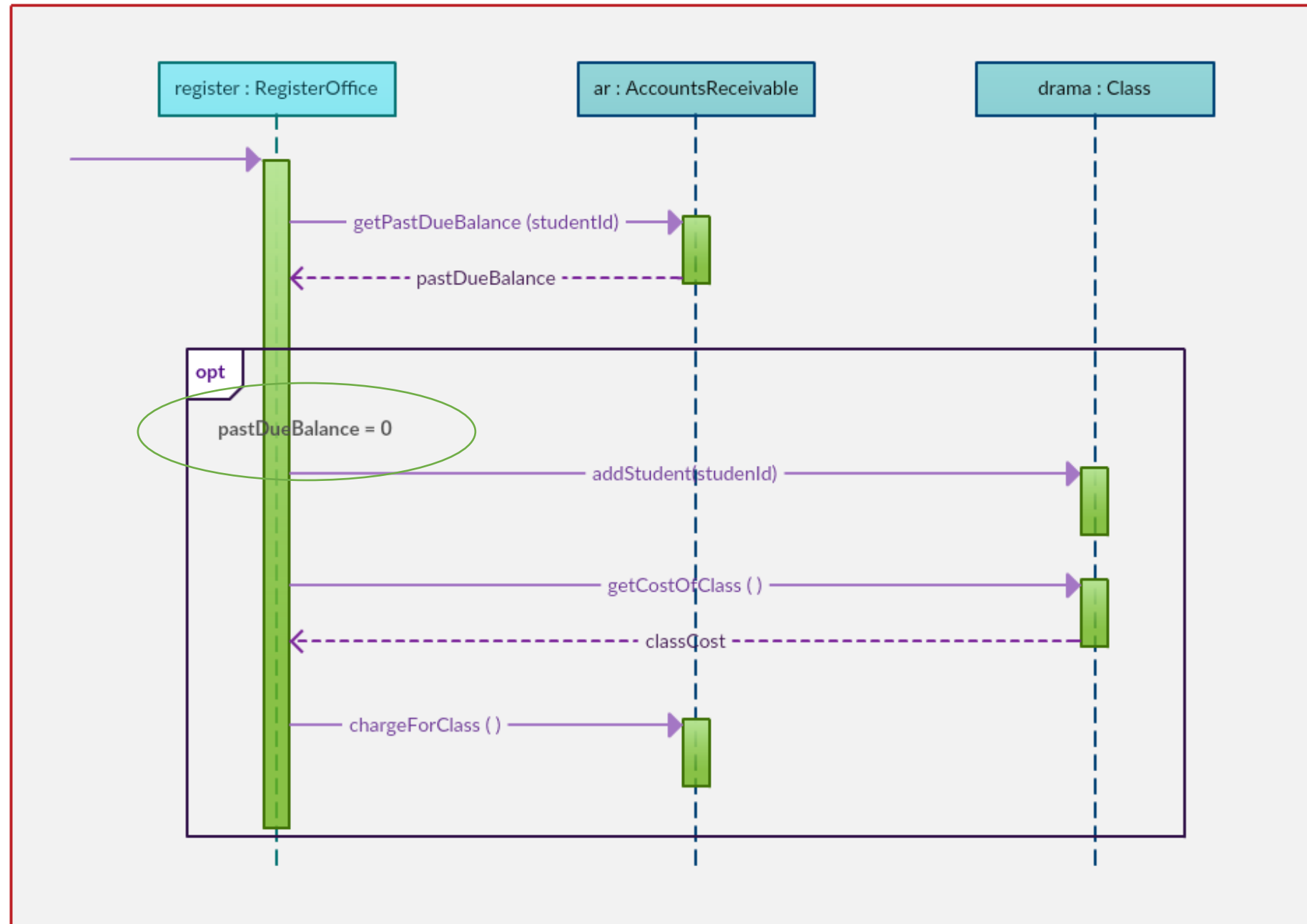
- Alternatives (“alt”) allow the modeling of the “if then else” / “switch” logic





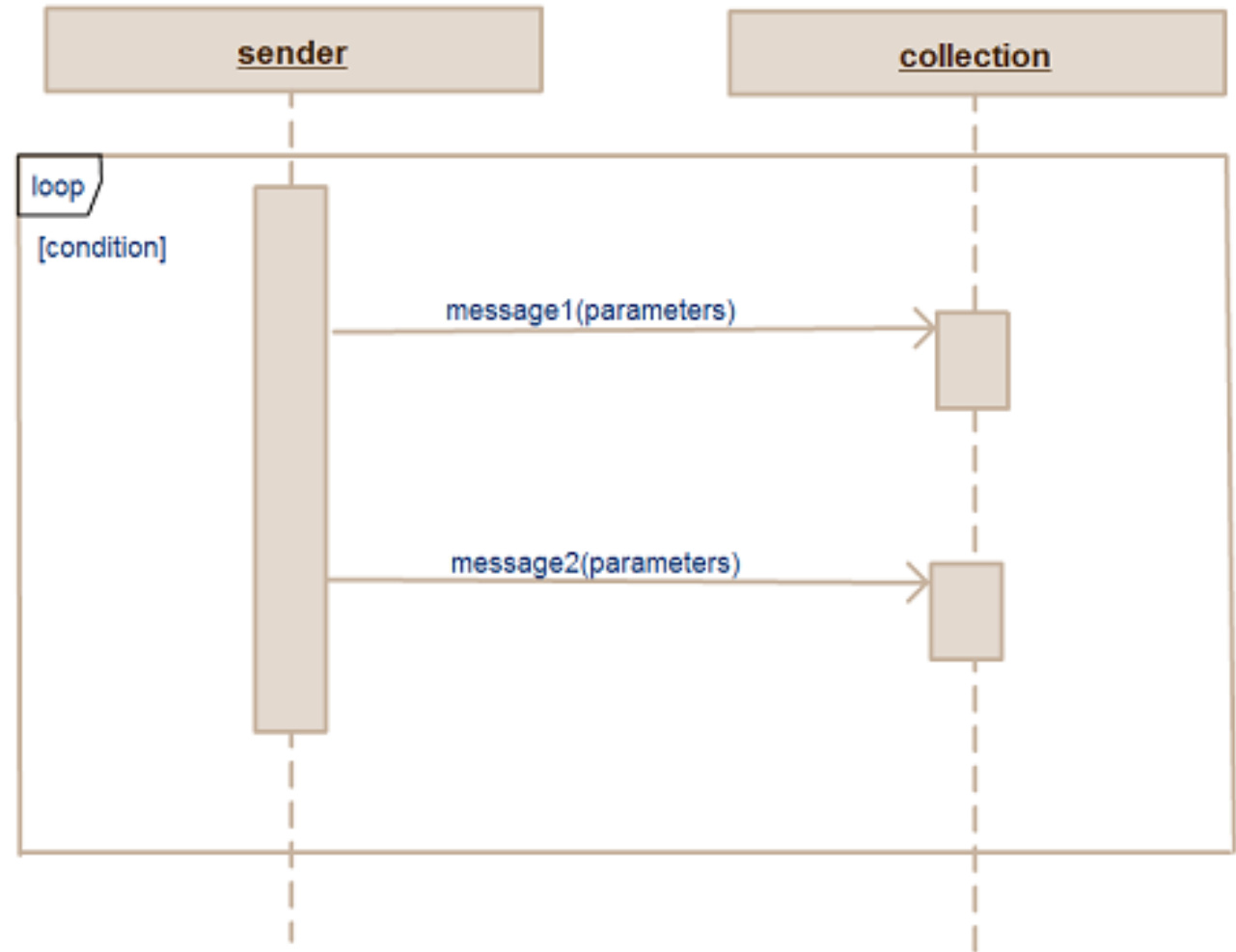
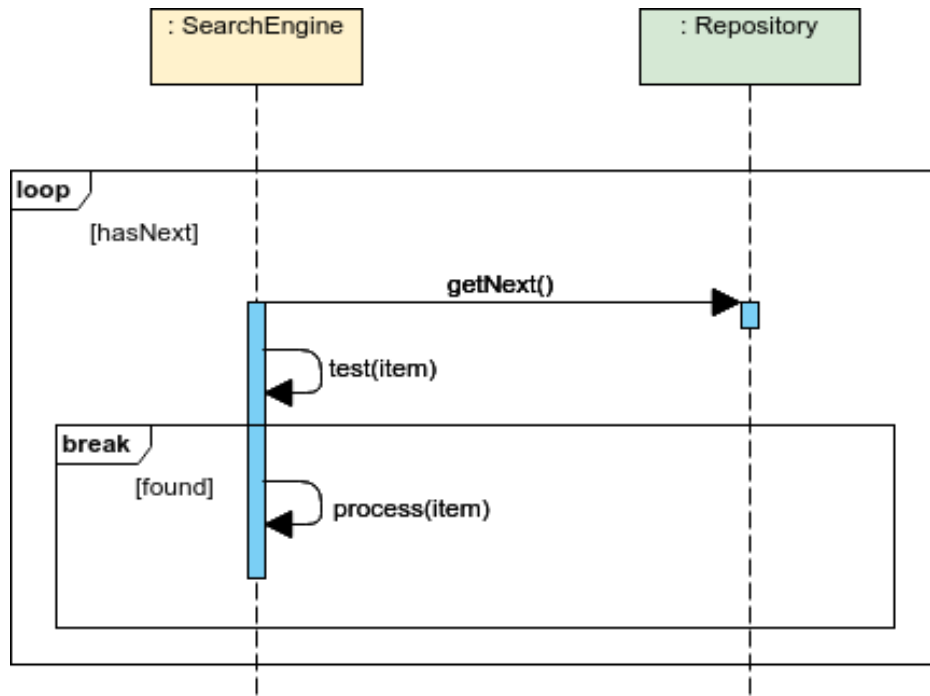
5. Option

- An option (“opt”) is usually used to model a simplistic “if then” (if without else) statement



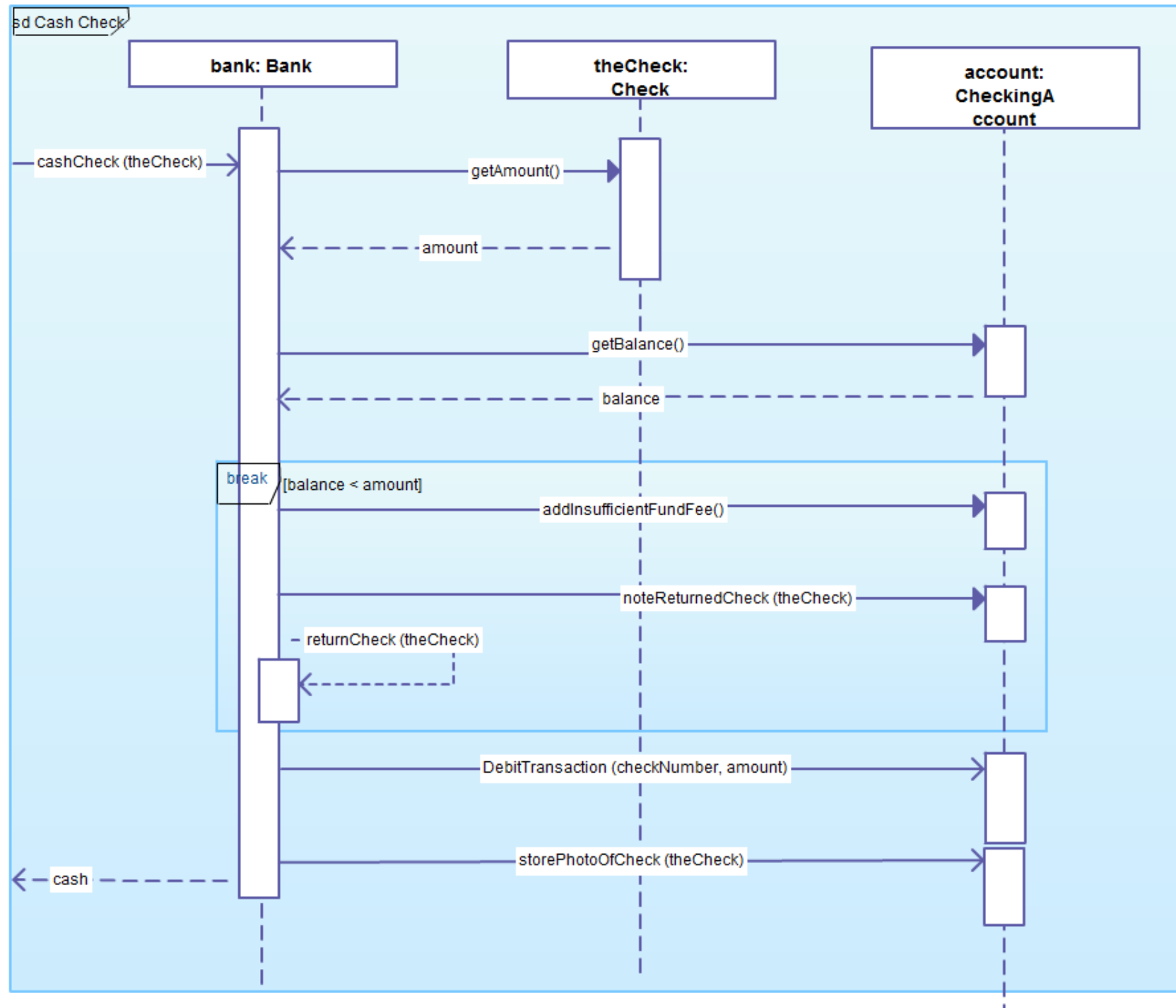
6. Loops

- This is something that is used when you need to model a repetitive sequence



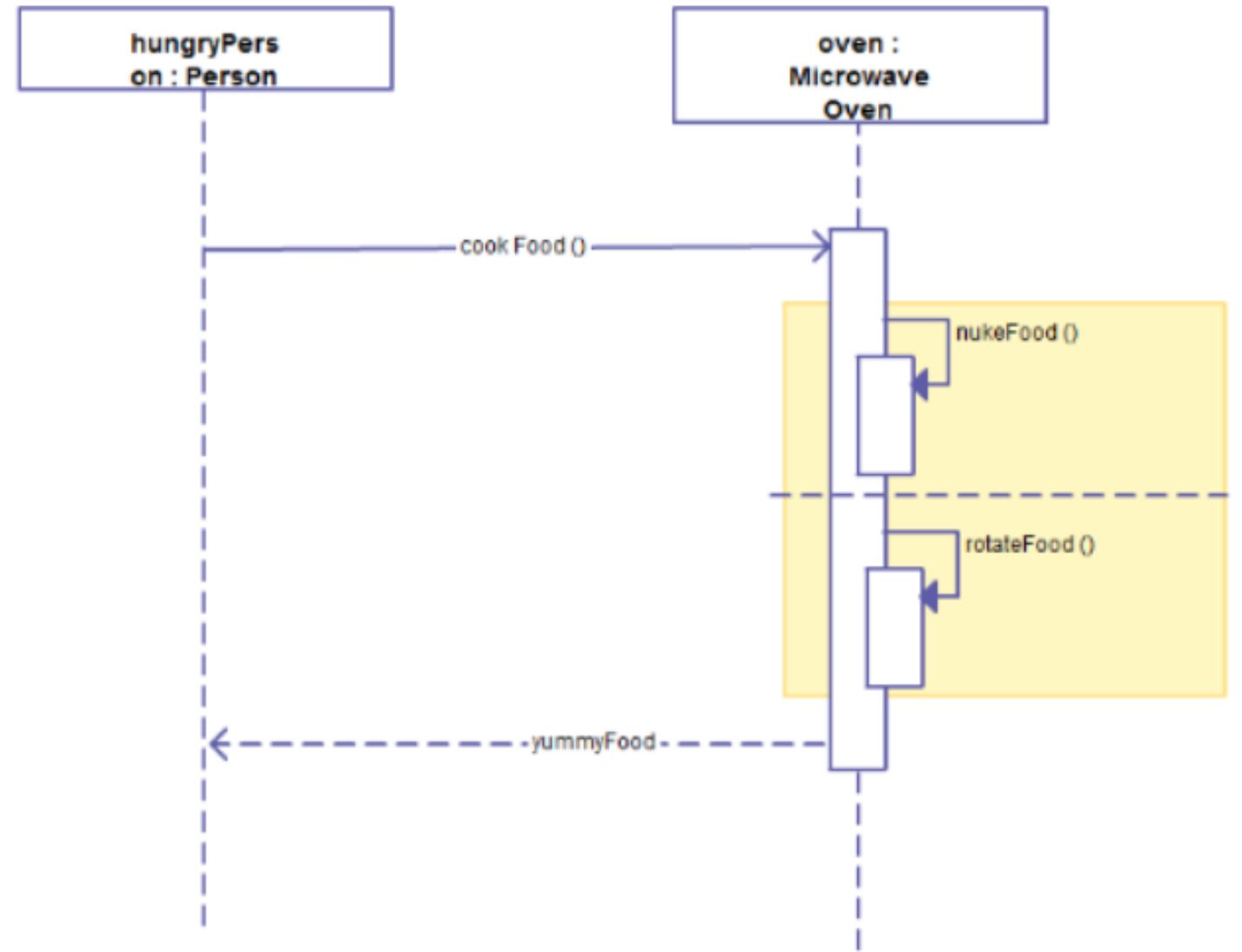
7. Break

- a break's frame has a name box stating "break".
- Breaks are utilized to model exception handling.



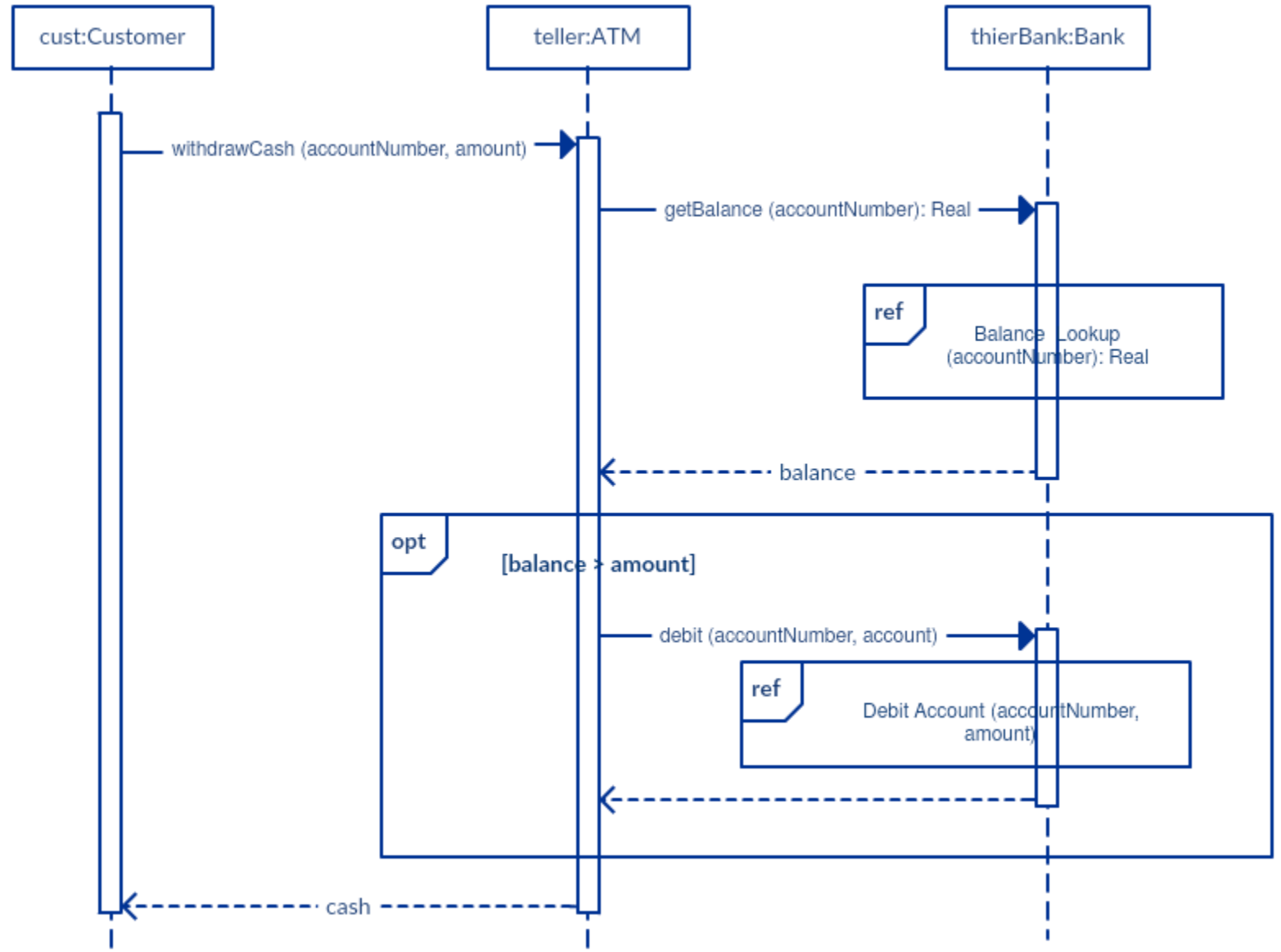
8. Parallel

- A hungryPerson sends the cookFood message to the oven object.
- When the oven object receives that message, it sends two messages to itself at the same time (nukeFood and rotateFood).
- After both of these messages are done, the hungryPerson object is returned yummyFood from the oven object.

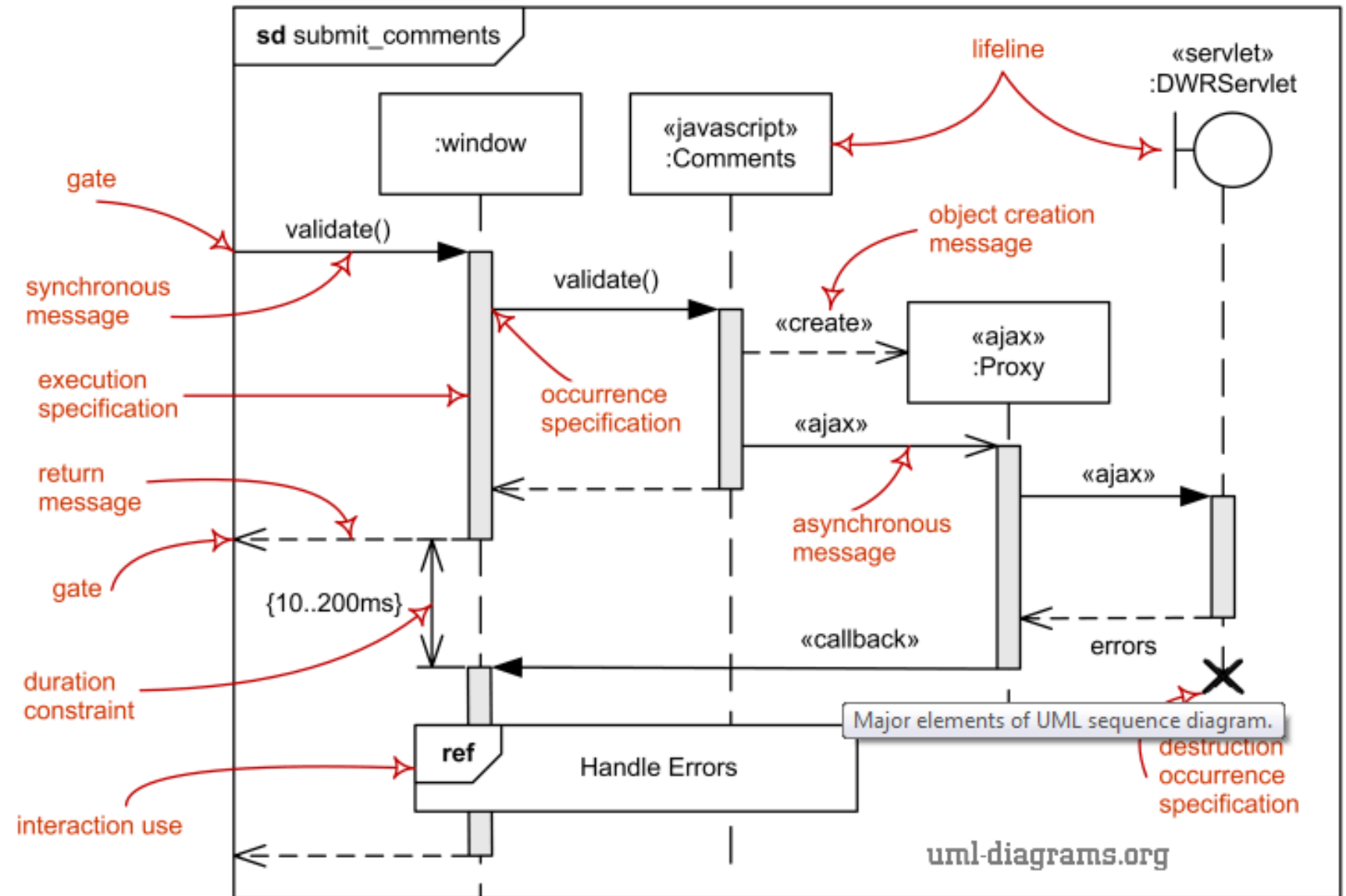


9. Reference Fragment

- use the ref fragment to manage the size of large sequence diagrams.
- It allows to reuse part of one sequence diagram in another, or in other words
- can reference part of a diagram in another diagram using the ref fragment.
- have to mention '**ref**' in the name box of the frame and the name of the sequence diagram that is being referred to inside the frame.



Example



How to Draw a Sequence Diagram

- need to draw the use case diagram and ready a comprehensive description of what the particular use case does.
- necessary to identify the objects (class) or actors.
- important to write a detailed description on what the use case does.
 - → easily figure out the interactions (that should go in the sequence diagram) that would occur between the objects above, once the use case is executed.

How to Draw a Sequence Diagram

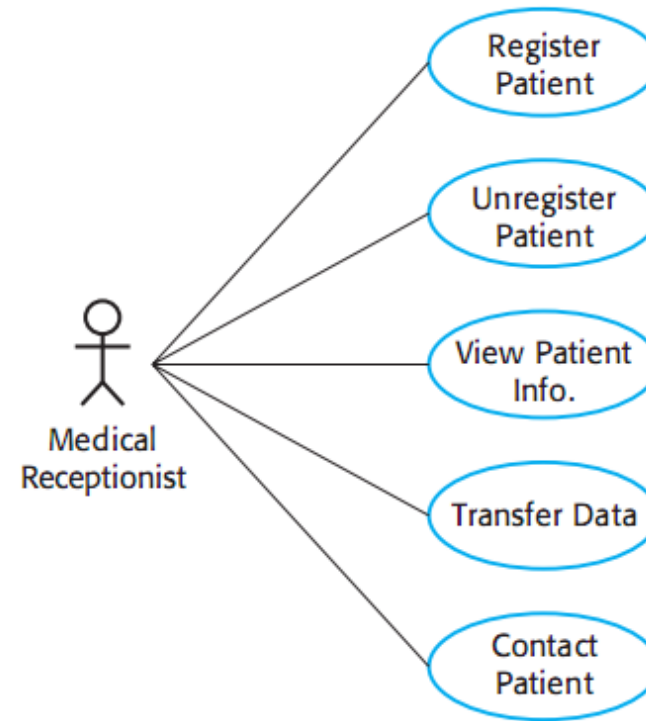


Figure 5.5 Use cases involving the role 'medical receptionist'

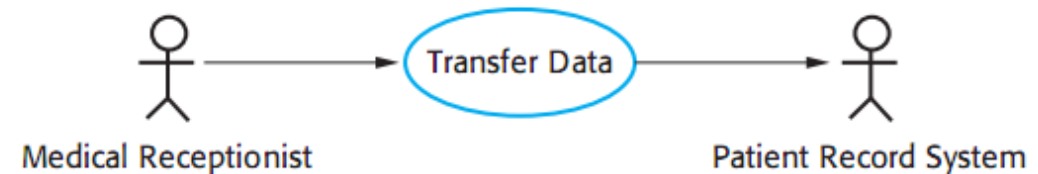
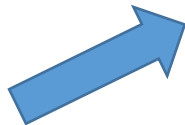


Figure 5.3 Transfer-data use case

How to Draw a Sequence Diagram



MHC-PMS: Transfer data	
Actors	Medical receptionist, patient records system (PRS)
Description	A receptionist may transfer data from the MHC-PMS to a general patient record database that is maintained by a health authority. The information transferred may either be updated personal information (address, phone number, etc.) or a summary of the patient's diagnosis and treatment.
Data	Patient's personal information, treatment summary
Stimulus	User command issued by medical receptionist
Response	Confirmation that PRS has been updated
Comments	The receptionist must have appropriate security permissions to access the patient information and the PRS.

In-class activity: draw a sequence diagram

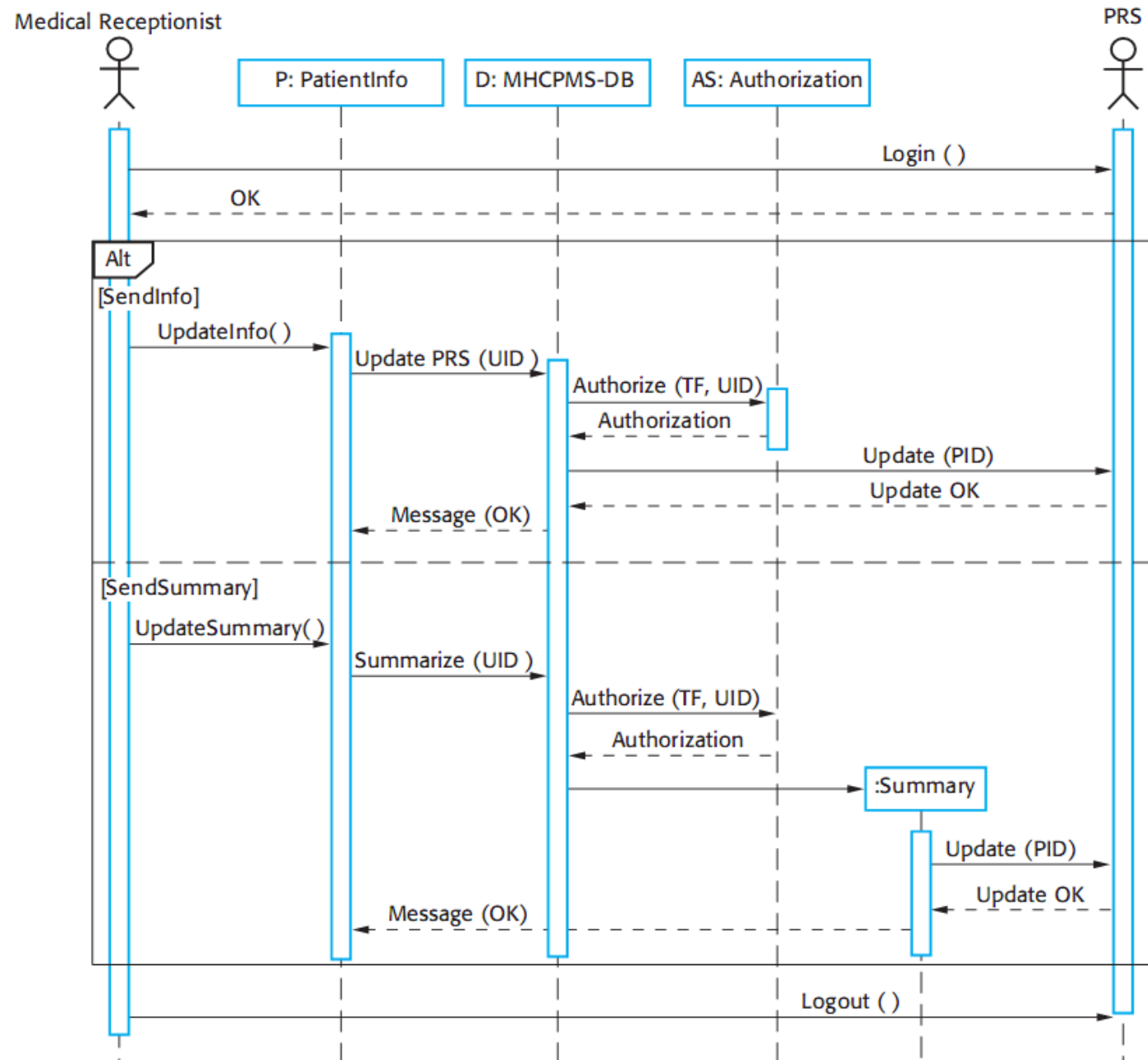
Note:

1. If in seq. diagram A ---- method1(xxx) ----> B, then "method1" is of A or B?

How to Draw a Sequence Diagram

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How to Draw a Sequence Diagram



Note [1]

- Không cần vẽ cho từng use-case
 - Một sequence diagram có thể cho 1 hoặc nhiều use-case
 - Cho nhiều use-case: phức tạp quá, không đọc được!
 - Sequence diagram không cho use-case nào?
 - Yes: miễn là giao tiếp nội!
- Biểu diễn tương tác bên trong của hệ thống
 - Cần các đối tượng bên trong hệ thống!
 - Và tương tác với nhau!
- Actor nhất quán từ use-case => sequence diagram!

Note [2]

- Động từ của tương tác?
 - Sai: “Bộ xử lý” --“Gửi đến server”--> “Server”
 - “Gửi...” là hành động của “Bộ xử lý”, không phải của “Server”
 - Đúng: A --”tác vụ”--> B thì “tác vụ” được định nghĩa trong/thuộc về B
- Tên tương tác có nghĩa và phân biệt (với các tương tác khác)
 - Sai: “get()” vì không rõ nghĩa
- Không có ”tác vụ mờ côi”
 - không được kích hoạt từ actor hoặc từ tác vụ khác
- Tuân thủ thứ tự thời gian

Submit Project #3

- Diagram
- Description of diagram